

Chapter 8 - Relations

4 pts

1. Let $A = \{a, b, c, d\}$ and $R = \{(a, a), (a, c), (b, d), (c, a), (c, c), (d, b)\}$ be a relation on A .

(a) Draw a digraph which represents R .

(b) Draw the bipartite graph which represents R .

2 pts

2. Let $A = \{a, b, c, d\}$ and $R = \{(a, a), (a, c), (b, d), (c, a), (c, c), (d, b)\}$ be a relation on A . What is the inverse of R ?

- 2 pts 3. Find the composition, $R \circ S$, where $S = \{ (1, a), (4, a), (5, b), (2, c), (5, c), (3, d) \}$ and $R = \{ (a, x), (a, y), (b, x), (c, z), (d, z) \}$ as a set of ordered pairs.

- 4 pts 4. (a) True False For any relation R , $(R^{-1})^{-1} = R$.

- (b) True False There are no relations R for which $R = R^{-1}$.

- (c) True False Let A and B be two sets. The empty set, $\emptyset$, is a relation from A to B .

- (d) True False Let S be a relation from A to B , and let R be a relation from B to C . Then

$$(R \circ S)^{-1} = S^{-1} \circ R^{-1}.$$

Chapter 9 - Properties of Relations

- 2 pts 1. Give a counterexample to the claim that a relation R on a set A that is both symmetric and transitive must be reflexive. Hint: There is a very simple example!

10 pts

2. Let A be the relation on the set $\mathbb{Z}$ of all integer defined by $s A t$ if and only if $|s| \leq |t|$.

(a) Is A reflexive? Justify your answer.

(b) Is A irreflexive? Justify your answer.

(c) Is A symmetric? Justify your answer.

(d) Is A antisymmetric? Justify your answer.

(e) Is A transitive? Justify your answer.

Chapter 10 - Equivalence Relations

6 pts

1. Let L be the collection of all straight lines in the plane. Four examples of elements in L : $x + y = 0$, $2x - y = -5$, $x = 7$, $y = 0$. A relation P on L is defined by the rule $l_1 P l_2$ provided the lines l_1 and l_2 are parallel. Is P an equivalence relation on L ? If it is, what are the equivalence classes of P ?

6 pts

2. Let A be the set of all ordered pairs of positive integers. So some members of A are $(3, 6)$, $(7, 7)$, $(11, 4)$, and $(1, 2981)$. A relation on A is defined by the rule $(a, b)R(c, d)$ if and only if $ad = bc$. For example $(3, 5)R(6, 10)$ is true since $(3)(10) = (5)(6)$.

(a) Explain why R is an equivalence relation on A .

(b) List four ordered pairs in the equivalence class of $(2, 3)$.